

EXAMPLE :

DESIGN LOGIC C

Digital Logic Design

A	B	X
0	0	1
0	1	0
1	0	1
1	1	1

POS

PRODUCT OF SUM

A	B	X
INVERT a variable with 1 input by adding (') to the equation.		
1	0	1
1	1	1

To determine the POS, look at the **X(output)** that has the value of 0 then write the equation using Addition (+) or using **OR LogicGate**.

1	1	1
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SOP

SUM OF PRODUCT

A	B	X	
INVERT a variable with 0 input by adding (') to the equation.			
1	0	1	$A*B$
1	1	1	$A*B$

To determine the SOP, look at the **X(output)** that has the value of 1 then write the equation using Multiplication (*) or using **AND LogicGate**.

1	1	1	$A*B$
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POS

The equation

$$X = A + B' \quad (\text{SIMPLIFIED})$$



**THEY HAVE THE
SAME
SIMPLIFIED
EQUATION**

SOP

The equation

$$X = A' * B' + \underline{A * B'} + A * B$$

$$X = A' * B' + \underline{A(B' + B)}$$

w/ the use of
DISTRIBUTIVE LAW

$$X = A' * B' + \underline{A}$$

w/ the us of
COMMON IDENTITY LAW

$$X = A + B' \quad (\text{SIMPLIFIED})$$



Logical Circuit Equivalent

